



# Lean Delivery & Value Stream Mapping

Week 7 • Day 4 • Agile Delivery & ADO Mastery

TPM Academy



Week 7 builds Agile delivery and Azure DevOps (ADO) mastery.

## Day 4 Objectives

- Identify the **value stream** from idea to value-realized (7–10 steps)
- Measure or estimate **process time**, **lead time**, and **flow efficiency** per step
- Find the **top 3 queues** with named cause + candidate intervention
- Draw the **visual VSM**
- Use AI to surface **non-obvious queues**



# Today's Shape

| Clock         | Block                       | Outcome                       |
|---------------|-----------------------------|-------------------------------|
| 09:00 – 09:15 | Opening                     | Pin DP Sections 1–3           |
| 09:15 – 10:30 | Block 1 + Activity 1        | Identify your value stream    |
| 10:45 – 12:00 | Block 2 + Activity 2        | Measure or estimate each step |
| 13:00 – 14:15 | Block 3 + Activity 3        | Top 3 queues                  |
| 14:30 – 16:00 | Block 4 + Activity 4 + Wrap | VSM + AI critique             |

# Block 1 — The Lean Claim

Most of the time, work is waiting



# Lean's Foundational Claim

Most of the time work spends in a system is **waiting**, not being worked on. Reduce waiting; reduce total time.

**Typical software example:** a story with 2 hours of actual coding may spend 3 weeks in the system — waiting for spec, review, QA, deploy slot, merge.

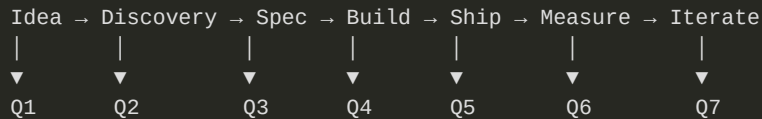
A TPM who optimizes coding time saves minutes. A TPM who reduces the waiting saves weeks.



# VSM Vocabulary

| Term              | Meaning                                |
|-------------------|----------------------------------------|
| Value stream      | Whole flow from idea to value-realized |
| Process time (PT) | Hands-on-work time at a step           |
| Lead time (LT)    | Total time at a step including waiting |
| Flow efficiency   | $PT / LT \times 100\%$                 |
| Queue             | Work waiting between steps             |
| Cycle time        | Lead time across the whole stream      |
| Throughput        | Completed items per unit time          |
| WIP               | Items currently being worked           |
| Little's Law      | $Cycle\ time = WIP / throughput$       |

# Typical Software Value Stream



## Worst queues for most teams:

- Spec → Build (waiting for engineering capacity)
- Build → Ship (review, QA, deploy slot)
- Ship → Measure (data accumulation; nobody looks)

# Activity 1 — Identify Your Stream

Quads • 45 minutes

- 15 min: walk steps idea → value-realized
- 10 min: identify queues between steps
- 10 min: mark step boundaries explicitly



15 + 10 + 10



# Block 2 — Measure or Estimate

Process time / lead time / flow efficiency



# Worked Numbers (FieldPulse)

| Step              | PT          | LT           | Flow eff |
|-------------------|-------------|--------------|----------|
| Idea              | 1 hr        | 3 days       | 5%       |
| Discovery         | 4 hrs       | 2 days       | 25%      |
| Spec              | 16 hrs      | 2 weeks      | 22%      |
| Estimate + accept | 1 hr        | 1 week       | 3%       |
| Build             | 8 hrs/story | 1 week/story | 14%      |
| Code review       | 30 min      | 1.5 days     | 4%       |
| QA + integration  | 4 hrs       | 2 days       | 12%      |
| Deploy            | 30 min      | 6 hrs        | 8%       |

**Total cycle time:** 6–8 weeks. **Total flow efficiency:** 5–15% typical.

## Activity 2 — Measure / Estimate

Quads • 50 minutes

- 40 min: PT + LT + flow eff per step



50 min straight

# Block 3 — Top 3 Queues

Find where the time hides

# Queue Investigation

For each top queue:

- **Why does this queue exist?** (capacity, policy, inertia, hidden dependency)
- **What would shrink it?** (capacity, policy change, parallel work, dependency removal)
- **Cost of shrinking?** (people-time, money, organizational change)

**"We're slow in code review"** is not specific. **"PRs wait 2 days for first review because we have 8 reviewers and 40 PRs/week"** is.



## Worked Top 3 Queues

| Queue                       | Cause                         | Intervention candidate                                    |
|-----------------------------|-------------------------------|-----------------------------------------------------------|
| Spec → Estimate (1 wk)      | Sprint planning bi-weekly     | Mid-cycle sync; mid-sprint pull                           |
| Build → Code review (1.5 d) | 5 reviewers; PRs wait         | Pair-review during dev; auto-assign                       |
| Discovery → Spec (varies)   | Customer interview scheduling | Async cycles; lighter-weight discovery for known patterns |

# Activity 3 — Top 3 Queues

Quads • 50 minutes

- 5 min: sum total queue time
- 10 min: identify top 3 queues
- 15 min: investigate each (cause / intervention / cost)
- 10 min: rank by impact / effort / risk



5 + 10 + 15 + 10

# Block 4 — Visual VSM + AI

One page; the queues hit emotionally





# The VSM Canvas

[Step 1]—→[Step 2]—→[Step 3]—→...

|

PT: \_\_

LT: \_\_

|

PT: \_\_

LT: \_\_

|

PT: \_\_

LT: \_\_

QUEUE

\_\_ items

\_\_ days

QUEUE

\_\_ items

\_\_ days

QUEUE

\_\_ items

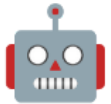
\_\_ days

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Total cycle time: \_\_ days

Total process time: \_\_ days

Total flow efficiency: \_\_%



# AI Critique Prompt

Role: Lean coach reviewing a software value stream.

Context: <description of the VSM with PT/LT/flow eff numbers and identified queues>

Task: Identify 3 queues likely missing or under-counted:

1. A queue between two steps not yet shown
2. A step where PT and LT seem out of line with team reality
3. An invisible queue (e.g., approvals, "ping someone")

Constraints:

- Use only the data given
- Be specific

Format: 3 numbered findings – what's missing / why suspect / how to verify.

## Activity 4 — VSM + AI

Quads • 55 minutes • Wrap

- 20 min: draw the VSM
- 10 min: AI critique
- 10 min: update VSM; recompute flow efficiency
- 5 min: document in Section 4



20 + 10 + 10 + 5 · 15-min wrap



# Day 4 Wrap

## What we built

- Value stream identified
- PT / LT / flow efficiency per step
- Top 3 queues with cause + intervention
- Visual VSM
- Delivery Plan (DP) Section 4 drafted

## What opens tomorrow

- **Bottleneck removal experiments**
- Hypothesis / test / success criterion
- Friday: full DP integration + sign-off

**Bring to Day 5:** DP Sections 1–4 + your top 3 queues. Tomorrow we design the experiments.

# Questions & Reflection

Where does your team's time hide that nobody talks about?